

● NIGHTPASS · DIGITAL BATTERY PASSPORT ON MIDNIGHT

The battery passport that proves *without revealing*

Every Passport anchored on Midnight, verified with zero-knowledge proofs. Integrates with any ERP solution, fully SAP-compatible by design.

18 Feb 2027

the passport becomes mandatory

7–9 M / yr

batteries needing one, from day one

Billions

of units/yr as more product passports follow

A deadline with liability attached

Mandatory passports for EV, e-bike and industrial batteries

EV and LMT batteries, plus industrial batteries > 2 kWh sold in the EU need one, with about 80 mandated attributes.

The operator is liable for the information inside the passport

Whoever places the battery on the market answers for the data: carbon footprint, origin, recycled content.

Disclosure conflicts with supplier NDAs & own trade secrets

Some of the information to be published in the passport is often subject to supplier NDAs or would disclose the company's own trade secrets.

18.02.2027 Reg. (EU) 2023/1542, Art. 77.

7 to 9 million passports a year, required by law

2–3 M

EV packs

~5 M

LMT • e-bikes & scooters

100k+

storage

By units: e-bikes lead

5.1 M e-bikes sold in Europe in 2023. Small batteries, the largest unit count.

By energy: EVs dominate

2.2 M electrified cars in 2024; JRC projects ~320 GWh/yr by 2030.

Fastest growth: storage

27.1 GWh installed in 2025, up 45 %. Twelfth record year in a row.

JRC • IEA • Fraunhofer ISI • CONEBI • SolarPower Europe

Batteries are just the first category. ESPR (Ecodesign for Sustainable Products Regulation) adds steel, textiles, tyres and aluminium which scales the demand to billions of passports per year.

Prove the claim, keep the number

AT ANCHOR TIME

1. Every provable field becomes a hashed leaf
2. The leaves fold into one Merkle root
3. The root is anchored with the passport

AT PROOF TIME

1. The circuit recomputes the leaf from the secret value
2. It must fold back to the anchored root, so the value comes from *this* passport
3. A false claim never lands as a transaction

Publicly visible: `carbon footprint ≤ 4000 kg CO2e: proven tx:0x83..7b`

One dataset, three audiences

level 0

Consumer / Public

Scan the QR, see the label
and basic information.
Anonymous, no account.

level 1

Recycler

Chemistry, capacity,
recycled shares, including
live telemetry.

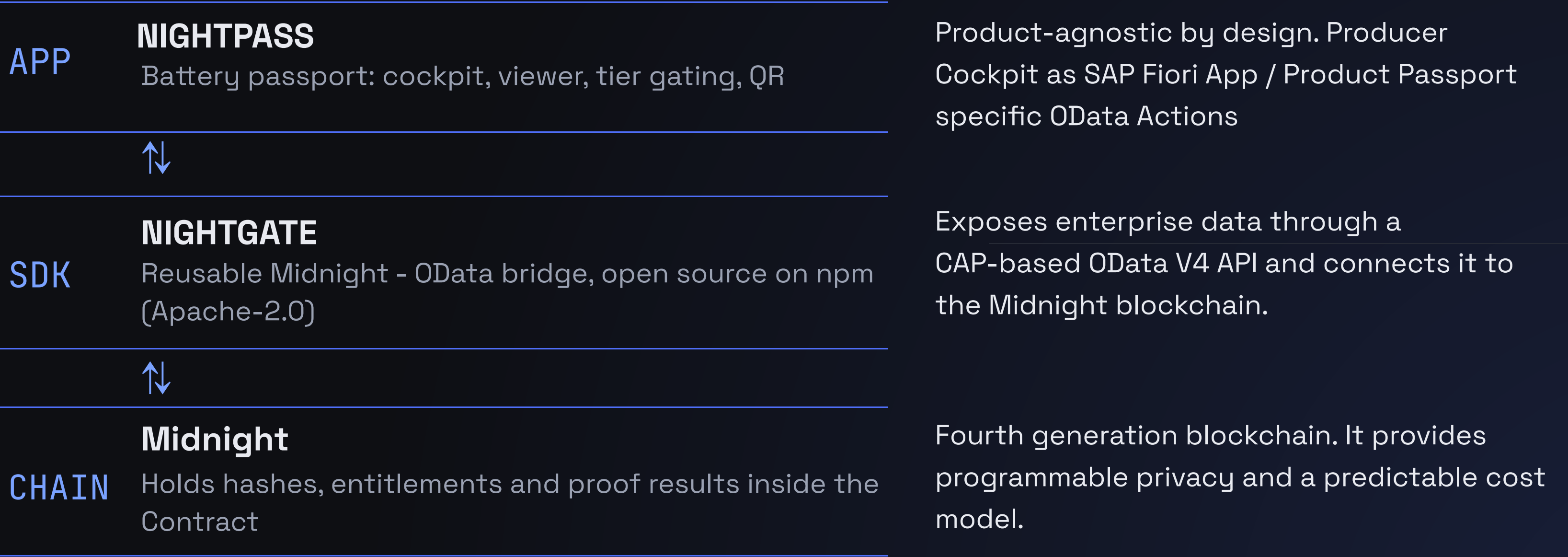
level 2

Authority

Everything in cleartext:
suppliers, carbon footprint,
due-diligence evidence.

An on-chain grant raises a partner's tier per passport and is revocable.

Three layers, one rule: sensitive data stays off-chain



No cryptographic complexity belongs to the user's workflow

Batched anchor transactions

attest → **bind** → **root** in a single tx,
deterministic order.

Shielded by default

No sender address, no cleartext arguments. Public
is the *what* and *when*, never the who.

Zero-infrastructure proving

ZK proofs can also run in-process to ensure the
highest level of confidentiality.

Sponsored DUST fees

A sponsor pool pays every fee; producers can
anchor passports from zero-funded wallets.

Batched ZK Proofs

Prove up to 8 passport field values in one
transaction

Verification is only a live ledger read

Anyone can verify the passport's integrity without
a wallet, an account or trust in a database.

Key achievements verifiable today

Live Public Demo with 55+ verifiable Passports on preprod

Mint a demo ZK battery passport at demo.zkpassport.eu in ~5 minutes on Midnight preprod, fees sponsored, batched circuit calls, verifiable on the public explorer

EU DPP Registry enrolled • 11/11 interoperability scenarios • 0 validation errors

- Enrolled in the EU DPP Registry Test Environment as a verified economic operator
 - Passed all 11 official BatteryPass-Ready interoperability scenarios.
 - Zero validation errors against the official Battery Pass consortium API
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End to End SAP Integration

Real S/4HANA goods receipts auto-minted into anchored, verified passports, live against SAP's API sandbox. SAP Fiori Producer Cockpit App

Two segments, one platform

ENTERPRISE

EV OEMs, importers, BESS makers

~2–3 M packs/yr · deep S/4 integration
“Passport-ready in 90 days”: fixed-price pilot,
then per-unit rollout

VOLUME SAAS

E-bike makers & importers

~5 M units/yr · multi-tenant, self-service
The largest unit count in the market, priced
per unit

Q3 2026

Launching Full Product
Version V1

Q4 2026

1–2 first pilot
integrations

Feb 2027

Production launch,
per-passport revenue

2027/28

ESPR schema #2: steel & more
product passport integrations



Maximilian Weber
Senior Software
Developer & Architect

The person behind NIGHTPASS

SAP & enterprise background

Degree from Darmstadt University of Technology / 10 years ERP Development & Consulting / Self-employed consultant for 4 years

Shipped, not just pitched

Contract, platform plugin, producer cockpit, public explorer and public demo, live on Midnight preprod today.

In the ecosystem

Midnight Build Club Cohort 1 / Certified Aiken Smart Contract Developer / Cardano Project Catalyst (Cardano-SAP integration), delivered three months early / Winner Charli3 Oracle Hackathon 2026

Thanks for listening
Questions?

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https://github.com/ODATANO

GitHub